## Google Glass – Moonshot Factory / Google X Project Failure

**Research:**

Google Glass was an innovative pair of wearable smart glasses that failed due to privacy concerns, price, technological limitations, and user confusion. Google Glass was intended to be an approachable and entry level device that would introduce Augmented Reality to consumers. However, users were unfortunately confused as to why they would want to use the technology themselves and the hefty price tag also strongly deterred any interest.

The Moonshot Factory, also called Googe X lab, is a collection of innovative Google projects. These projects include self-driving cars, molecular recycling, atmospheric water harvester and the like. Many projects have either become independent companies or were integrated with other Google products. Google Glass was a project that was a part of the Moonshot Factory repertoire. It launched the general public's Explorer Edition in 2014 and the commercial Enterprise Edition in 2017. The consumer public version failed quite quickly and was discontinued in 2015 due to privacy concerns and technical limitations. The enterprise version stayed in the market for about six years before being discontinued. Google never publicly gave a definitive reason as to why they ended production and support for enterprise version. Though many speculate that there was not enough demand for such a product and that Google needed to cut the product out of its portfolio to focus on more current and lucrative products.

Google Glass failed to entice consumers as people were confused as to why they would need to incorporate such a clunky device into their daily lives. The glasses weren't

discreet. They had a unique design, and people could tell that they were not normal prescription glasses. The new age robot aesthetic wasn't a fashion statement that the public was eager to follow. The device also had a short battery life and wasn't well suited to read long texts. The device also had a short list of voice commands, and the voice control feature was also difficult to execute. This was partially due to the limitations of technology at the time such as the dependence on phone or Wi-Fi data. Google glass had Wi-Fi and Bluetooth but there was no built in GPS or cellular connection. Most consumers didn't want to tether by using their phones because of the potential carrier charges. The prices at the time were even higher than they are in current times. People were also worried about privacy concerns and the possibility of being recorded without consent.

Google glass lacked the flexibility and consistency that people expect from a general-purpose device. It was genuinely awkward to use when it came down to long term usage. Users navigated the device by tapping or swiping the touchpad or tilting their heads to choose from short menus. This was fine for short and quick actions. However, it wasn't sustainable or practical for long time use. The app selection was also lacking and small. Users felt that the idea of hands-free interaction sounded intriguing but not if it entailed performing a list of undesirable and strange movements! Despite the Enterprise version having some level of success in the market, the product was an ultimate failure. More user testing at the research and development stage could have saved them a lot of time and money. Unfortunately, Google was too eager to be the first to launch such a product to the public. So eager that they didn't acquire enough user feedback to see that they product was simply awkward and uncomfortable. Not to mention expensive!

**Post Mortem Case Study:**

Google Glass failed due low user engagement, price, technological limitations, and privacy concerns.

Issues:

- User confusion on the purpose and necessitation of the wearable
- Privacy concerns on being recorded without consent
- Awkward physical executions that weren't practical for long term use
- Poor voice command and control
- The need to tether to a mobile device meant having to use carrier fees
- Design was clunky and not practical
- Limited cross-platform capability: Users could only text Android users

Lessons:

- Early user feedback is crucial
- More usability testing in R&D
- Technological adaptability is key
- Cross platform usage is essential
- Design must be unobtrusive and practical
- Physical commands must be practical for long term usage
- Strong voice control is essential for hands free use